SEMICONDUCTOR PHYSICS AND THE HALL EFFECT

GENERAL DESCRIPTION

The Hall effect is used very commonly to characterize semiconductor samples and to investigate new phenomena, because it allows one to separately determine (along with measurements of the resistance) the carrier concentration, the sign of the carrier charge, and the scattering time of carriers. Resistance measurements alone only determine the product of the carrier concentration and scattering time. The temperature dependence of these quantities provides even more information, and can be used to deduce the energy bandgap of the semiconductor and the characteristics of electrically active impurities: donor or acceptor, dopant concentration, and binding energy.

In this experiment you will investigate the temperature dependence of the carrier concentration and mobility in a sample of lightly doped p-type Ge. The sample is a single crystal of Ge doped with acceptors in a concentration $N_A = 2x10^{11}\,cm^{-3}$. At room temperature the sample is intrinsic: the concentrations n and p of electrons and holes are equal and determined by the generation of electronhole pairs across the band gap. At lower temperatures the carrier concentration decreases exponentially in an activated fashion, until it approaches the acceptor concentration. At lower temperatures the sample is extrinsic and ptype. The concentration of holes is much greater than electrons and relatively temperature independent $p = N_A$. The mobility of electrons and holes is also temperature dependent throughout this range.

The geometry of the sample is a square with soldered indium contacts on each comer. Using the van der Pauw technique of exchanging leads and averaging resistance and Hall measurements, one can obtain accurate measurements of the resistivity and Hall constant even for non-ideal geometries such as this. The sample is heat sunk to a copper cold finger near a diode thermometer, which is read out on the Lakeshore temperature controller.

APPARATUS

Liquid nitrogen cryostat with copper cold finger
Lakeshore Cryotronics temperature controller
Vacuum pump and gauge
Ge sample - leads brought out to connection box
Electromagnet and power supply

Gaussmeter
Lockinamplifier with built-in oscillator
WaveTek function generator
Series resistors in Pomona boxes
Oscilloscope
Multimeter

PROCEDURE

Before you start, check that the sample is mounted in the cryostat. Ask the staff for help to open the cryostat. Remount the cryostat on its stand and pump out the vacuum jacket to about 10 microns on the thermocouple vacuum gauge. Valve off the pump and fill the cryostat with liquid nitrogen. The sample will cool with a time constant of about one half hour. As the sample cools, you can control the temperature using the Lakeshore temperature controller. A power supply can also be used to heat the cold finger manually. The lowest temperature achievable with liquid nitrogen is about 85K. It is possible to fill the precooled cryostat with liquid helium to cool to about 20K. Consult the staff if you would like to do this.

Make electrical measurements using the sinusoidal voltage source at the back of the lockin amplifier, which has a frequency of 1 kHz and a switchable value of 1 Vrms to 0.01 Vrms. Connect the reference out on the back panel to the reference input on the front panel. Alternatively, use the Wavetek function generator, connecting its SYNC output to the lock-in reference. Measure the voltage source using a multimeter, and convert it to a current source using a large series resistor in a Pomona box of typically 100k to 1M. The value of this resistor should be much larger than the resistance of the sample, which is strongly temperature dependent. Connect the Pomona box to one current lead, and complete the circuit at the other current lead using a shorting cap or the equivalent. The voltage is measured differentially using both inputs of the lockin amplifier. To do the van der Pauw technique, you will need to switch leads at the sample connector box. Adjust the phase of the lockin and verify that everything is working correctly by monitoring the signal at the monitor output connector on the front panel.

The electromagnet is powered by a large d.c. power supply. Never disconnect the leads when current is flowing, or you will generate a large, dangerous and destructive voltage, just as you do for the spark plugs in your car. The magnetic field at the sample is the sum of that due to the coils and due to the magnetization of the iron core, which creates a small hysteresis as one sweeps the current up and down. To accurately determine the magnetic field use a gaussmeter.

First measure the magnetoresistance and Hall resistance vs. magnetic field (both polarities) at the lowest and highest temperatures, about 85K and 300K. The magnetoresistance is even in the applied resistance and the Hall effect is odd, but nominal measurements are usually a combination of both. One can separate the two effects using their symmetry i.e. the Hall resistance is antisymmetric as one reverses the magnetic field, while the magnetoresistance is symmetric. Your Hall measurements should be taken at as low a magnetic field as possible without losing too much signal so that $\omega_c \tau << 1$.

Once you have determined the best magnetic field range, measure the resistivity and Hall constant vs. temperature, and use this data to determine both the carrier concentration and mobility vs. temperature. Actually compute and plot your results as you take the data; this saves much grief later if you make any mistakes. Note that at room temperature the number of electrons and holes is equal and one might think that the Hall resistance is zero (is it?). You can find a correct treatment of this case in the references.

REPORT

Your report should include high quality plots of both your raw data and quantitative plots of the carrier concentration and mobility vs temperature. Use these to determine the energy bandgap for Ge, the intrinsic carrier concentration at 300K, the type and concentration of impurities, and the mechanism, which limits the mobility. Be sure to include a careful discussion of the limitations of this technique and the sources of error in your measurements.

REFERENCES

R.B. Adler, A.C. *Smith,* and R.L. Longini, *Introduction to Semiconductor Physics,* (John Wiley & Sons, N.Y., 1964).

R.A. *Smith, Semiconductors,* (Cambridge Univ. Press, Cambridge, 1978).

D.A. Fraser, *The Physics of Semiconductor Devices,* (Clarendon Press, Oxford, 1979).

R.A. Greiner, *Semiconductor Devices and Applications, (McGraw* Hill, N.Y., 1961).

J. Seymour, *Electronic Devices and Components,* (John Wiley & Sons, N.Y., 1981).

N.W. Ashcroft and N.D. Mermin, *Introduction to Solid State Physics,* (Holt, Rinehart & Winston, N.Y., 1976).

R.K. Willardson and A.C. Beer, editors, *Semiconductors and Semimetals*, Vol 15 - Contacts, Junctions, Emitters, (Academic Press, N.Y., 12981).

http://icg.harvard.edu/~phys191r/pdf/d1.pdf